Database Development with PL/SQL INSY 8311

ADVENTIST UNIVERSITY OF CENTRAL AFRICA

Instructor:

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6h00 pm – 8h50 pm

- Monday A -G209
- Monday A -G209
- Tuesday B-G209
- Wednesday E-G209
- Thursday F-G308

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Database Development with PL/SQL

Reference reading

- What is PL/SQL?
- Oracle PL/SQL Articles
- Difference Between SQL and PLSQL
- 4 Oracle Database Installation and Configuration for Advanced Management Console
- Oracle SQL Developer Extension for VSCode
- Oracle DBA Tutorials: How to install Oracle Database 19c on Windows 11

Lecture 01 - Oracle Installation and Setup

What is PL/SQL?

What?

PL/SQL is a procedural relational database language developed by **Oracle Corporation** in earlier 90's.

PL/SQL – Procedural Language Structured Query Language.

Procedural language?

"PL/SQL allows the execution of a block of code at a time which increases it performance."

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Course Overview

Starting out with

Oracle

Covering Databases, SQL, PL/SQL, Developer Tools and DBA

This course provides an introduction to Oracle's database environment, including SQL, PL/SQL, and SQL*Plus. You will learn the fundamentals of these tools and languages, how they interact, and their applications in managing and manipulating relational databases.

Requirements for Oracle Database Installation

Before installing Oracle Database, ensure the following:

1. **Operating System:**
   - Windows 10/11, or compatible Linux/Unix versions.

2. **System Requirements:**
   - **Processor:** Minimum 2 GHz CPU
   - **RAM:** At least 8 GB (16 GB recommended)
   - **Disk Space:** Minimum 10 GB free space

3. **Privileges:**
   - Administrator privileges on your system.

4. **Software Requirements:**
   - **JDK:** Oracle JDK 8 or higher
   - **.NET Framework:** Version 4.5 or later (for Windows)
   - **Web Browser:** Updated version for accessing Oracle tools

5. **Environment Configuration:**
   - **Internet Connection:** For downloading installation files
   - **Unzip Utility:** Software like WinRAR or 7-Zip

6. **Optional:**
   - **Firewall/Antivirus:** Temporarily disable during installation (remember to re-enable afterwards)

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Setting Up Oracle Database 21c and SQL Developer: A Beginner's Guide (1/3)

Step 1: Download Oracle Database 21c

a. **Visit the Oracle Downloads page**: Oracle Database 21c Download
b. **Choose the appropriate version** for your operating system (Windows/Linux).
c. **Sign in to your Oracle account** or create a new one to download the software.
d. **Download the installation files** and save them to a location on your computer.

Step 2: Download SQL Developer 23.1.1

a. **Visit the Oracle SQL Developer page**: SQL Developer 23.1.1 Download
b. **Select the version** compatible with your operating system (Windows, macOS, Linux).
c. **Sign in to your Oracle account** or create a new one to download the software.
d. **Download the ZIP file** and extract it to a location on your computer.

Step 3: (Optional) Download Oracle SQL Developer Extension for VSCode

a. **Visit the Visual Studio Marketplace**: Oracle SQL Developer Extension for VSCode
b. **Click on "Install"** or open Visual Studio Code and search for "Oracle SQL Developer" in the Extensions marketplace.
c. **Install the extension** directly from within VSCode.

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Setting Up Oracle Database 21c and SQL Developer: A Beginner's Guide (2/3)

Step 4: Install Oracle Database 21c

a. **Navigate to the folder** where you downloaded the Oracle Database 21c installation files.
b. **Run the installer** and follow the on-screen instructions:
   - Accept the license agreement.
   - Choose the installation type (Desktop Class or Server Class).
   - Configure the database options as per your requirements (like system password).
c. **Wait for the installation to complete**. This might take some time.
d. **Once installed**, use the Oracle SQL*Plus or SQL Developer to connect to your new database.

Step 5: Install SQL Developer 23.1.1

a. **Extract the ZIP file** you downloaded for SQL Developer.
b. **Navigate to the extracted folder** and run sqldeveloper.exe (for Windows) or the equivalent executable for your OS.
c. **Start SQL Developer**, and if prompted, configure your JDK path.
d. **Connect to your Oracle Database** by creating a new connection in SQL Developer.

Optional Step: Configure SQL Developer Extension in VSCode

a. **Open Visual Studio Code**.
b. **Go to the Extensions tab** (Ctrl+Shift+X) and search for "Oracle SQL Developer".
c. **Install the extension**.
d. **Follow the instructions** provided by the extension to set up the connection to your Oracle Database.

Setting Up Oracle Database 21c and SQL Developer: A Beginner’s Guide (3/3)

Step 6: Follow a YouTube Tutorial for Installation (Optional)

For a visual guide on installing Oracle Database 21c, you can watch a YouTube tutorial:

- Access Oracle 21c from Google Drive
- **YouTube Tutorial:** Oracle Database 21c Installation Guide (Search for "install oracle database 21c")

This tutorial will walk you through the process step by step, ensuring that you correctly set up the database on your system.

Accessing and Using Oracle Live SQL

1. Open Oracle Live SQL
- **Go to Oracle Live SQL Website**: Visit Oracle Live SQL.

2. Start Coding
- **Click on "Start Coding Now"**: This will direct you to the login page if you don't already have an account.

3. Create an Oracle Account (If Needed)
- **Sign Up**: If you do not have an Oracle account, click on the sign-up option and follow the instructions to create a new account.
- **Log In**: If you already have an Oracle account, use your credentials to log in.

4. Access Oracle Live SQL Interface
- **After Logging In**: You will be redirected to the Oracle Live SQL interface.
  - **Overview**: The interface provides a space where you can write, execute, and save SQL and PL/SQL code.

5. Write and Execute SQL/PLSQL Code
- **SQL Worksheet**: You'll see an area to write SQL and PL/SQL code. Enter your code in the provided editor.
- **Run Code**: Click the Run button or use the appropriate keyboard shortcut (typically Ctrl+Enter or F5) to execute your code.
- **View Results**: Results will be displayed in the output pane below the editor.

6. Save Your Work
- **Save Scripts**: You can save your SQL and PL/SQL scripts by clicking on the Save button. You may need to provide a name for your script and select a location within your Live SQL account.

7. Use Additional Features
- **Script Library**: You can access and manage your saved scripts from the Script Library tab.
- **Example Scripts**: Explore example scripts provided by Oracle to understand various functionalities and use cases.
- **Documentation**: Refer to the documentation and help resources available within the interface for additional guidance.

Accessing and Using Oracle Live SQL

- **No Installation Required**: It’s a cloud-based service, so you don’t need to install any software.
- **Instant Access**: Quickly write and test SQL and PL/SQL code in a web-based environment.
- **Collaboration**: Share your scripts with others by saving them in your account.

Using Oracle Live SQL is a great way to practice SQL and PL/SQL, explore new queries, and experiment with code in an easy-to-access environment.

Assignment I: Installing Oracle Database and Executing SQL Commands

This assignment will be shared via the group WhatsApp. The rubric will outline the steps for completing and delivering the assignment. Please ensure you read and follow all instructions provided.

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Thank you!